Project 12 — Literature Review Papers (Simple Explanation + Links)

> All 6 papers from the Literature Survey table, with direct links, easy-to-understand findings, and the research gap each one leaves behind.

[1] SHAP — The "Score Every Feature" Paper

📄 **Full Title:** A Unified Approach to Interpreting Model Predictions 👨‍🔬 **Authors:** Scott M. Lundberg & Su-In Lee 📅 **Year:** 2017 | 🏛️ **Published at:** NeurIPS (Neural Information Processing Systems)

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/1705.07874
🏛️ Official NeurIPS Page	https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

🔬 What Did They Research?

Before this paper, there were many different tools that tried to explain AI decisions — but they all worked differently and gave different answers for the same model. It was a mess. These researchers asked:

> *"Can we create ONE unified, mathematically consistent framework to explain any AI model?"*

💡 What Did They Find / Build?

They built **SHAP** (SHapley Additive exPlanations).

The core idea — a simple analogy:

> Imagine 5 friends did a group project together. At the end, you want to fairly distribute the credit. You calculate: *"How much did each person contribute by comparing every possible combination of team members with and without that person?"* That's exactly what SHAP does — but for features in an AI model.

So for a loan rejection, SHAP would say:

- ✅ Your age contributed **+12%** toward approval
- ❌ Your low income contributed **-34%** toward rejection
- ❌ Your credit score contributed **-21%** toward rejection

It's clean, mathematically fair, and works on **any** ML model (XGBoost, Neural Net, etc.).

🚧 Research Gap (What's Missing?)

It only shows *correlation*, not *cause*.

SHAP tells you which features were *associated* with the rejection. But if income and savings are highly correlated (as they usually are), SHAP might give savings a high score just because it moves with income — not because savings itself was the actual cause.

> **Simple analogy:** SHAP might say "your shoes caused you to trip" when actually it was the wet floor. Both were present, and shoes are correlated with your movement — but the real cause is the floor.

In short: SHAP answers *"what was important?"* but not *"what actually caused it?"*

[2] LIME — The "Explain One Decision at a Time" Paper

📄 **Full Title:** "Why Should I Trust You?": Explaining the Predictions of Any Classifier 👨‍🎨 **Authors:** Marco Tulio Ribeiro, Sameer Singh, Carlos Guestrin 📅 **Year:** 2016 | 🏛️ **Published at:** KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining)

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/1602.04938
🏛️ ACM Official Page	https://doi.org/10.1145/2939672.2939778

🔬 What Did They Research?

AI models like deep neural networks are complex black boxes. You can't look inside and understand them. These researchers asked:

> *"Even if we can't understand the whole model, can we at least explain one specific prediction in a simple way?"*

💡 What Did They Find / Build?

They built **LIME** (Local Interpretable Model-agnostic Explanations).

The core idea — a simple analogy:

> You don't understand how a complex machine works overall. But you poke it in a small area, observe how it reacts, and draw a simple sketch of *that local region*. That sketch isn't the full machine — but it tells you about this one spot.

LIME does the same thing:

1. Take one input (e.g., your loan application)
2. Create many slightly changed versions of it (jiggle the values a bit)
3. See how the AI's output changes
4. Fit a **simple, understandable model** (like a straight line) to that small neighborhood
5. Use that simple model to explain **just this one prediction**

So LIME might say: *"For your specific case, income and employment status were the two most influential features."*

🚧 Research Gap (What's Missing?)

The explanations are unstable and still not causal.

- **Unstable:** Run LIME twice on the same input, and you might get different feature importance rankings. That's a trust problem.
- **Local only:** The explanation is only valid near that one data point. It doesn't generalize.
- **Still correlational:** Like SHAP, it captures statistical patterns — not real cause-and-effect.
- **No recourse:** It tells you *what mattered* but still gives you no guidance on *what to change*.

> **Simple analogy:** LIME is like a doctor saying "your blood pressure was high during this visit." It's a local observation. It doesn't tell you *why* it was high, and it doesn't tell you *what to do about it*.

[3] Actionable Recourse — The "What Can I Change?" Paper

📄 **Full Title:** Actionable Recourse in Linear Classification 👨‍🔬 **Authors:** Berk Ustun, Alexander Spangher, Yang Liu 📅 **Year:** 2019 | 🏛️ **Published at:** ACM FAccT (Fairness, Accountability, and Transparency)

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/1809.06514
🏛️ ACM Official Page	https://doi.org/10.1145/3287560.3287566
💻 GitHub Code	https://github.com/ustunb/actionable-recourse

🔬 What Did They Research?

This paper was one of the **first** to formally ask:

> *"If an AI rejects someone, what specific actions can that person realistically take to get a different outcome?"*

This is the concept of **algorithmic recourse** — giving people a path forward, not just a verdict.

💡 What Did They Find / Build?

They built a system using **Integer Linear Programming (ILP)** — a type of mathematical optimization — to find the *cheapest* set of changes to a person's features that would flip a model's decision from negative to positive.

Simple analogy:

> Imagine a GPS. You missed your destination. It doesn't just say "you're at the wrong place." It says "turn left in 200m, then take the highway for 5km." That's recourse — a navigation plan to get where you want to go.

Their system would output something like:

- Increase monthly income by ₹8,000
- Reduce existing debt by ₹15,000

And it specifically **respected constraints** — it wouldn't suggest changing your age or nationality.

🚧 Research Gap (What's Missing?)

It only works for simple (linear) models.

Linear models are the simplest kind of AI — they assume everything has a straight-line relationship. Most real-world AI systems (XGBoost, Neural Networks) are non-linear and far more complex.

- ❌ Doesn't scale to modern black-box ML models
- ❌ Doesn't consider **causal relationships** between features (e.g., if you increase income, your savings might also change — the model ignores this dependency)
- ❌ No user-facing dashboard or interface

> **Simple analogy:** This GPS only works on straight roads. The moment you hit a roundabout or a hill, it breaks.

[4] Counterfactual Explanations — The "What If?" Paper

📄 **Full Title:** Counterfactual Explanations Without Opening the Black Box: Automated Decisions and the GDPR 👨‍🔬 **Authors:** Sandra Wachter, Brent Mittelstadt, Chris Russell 📅 **Year:** 2017 | 🏛️ **Published at:** Harvard Journal of Law & Technology

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/1711.00399

🔬 What Did They Research?

This paper came from a **legal angle** — the EU's **GDPR law** (data protection law) says people have a right to an explanation when an automated system makes a decision about them. The authors asked:

> *"What's the simplest useful explanation we can give someone when AI rejects them — without needing to reveal or understand the model's internals?"*

💡 What Did They Find / Build?

They proposed **Counterfactual Explanations** — answers to the question: *"What would have had to be different for the outcome to change?"*

Simple analogy:

> You didn't get into a college. A counterfactual explanation is: *"If your score had been 5 marks higher, you would have been admitted."* It's the minimum change that flips the outcome.

Mathematically, they find the *closest possible input* (in feature space) to your original input, where the model gives a different output.

So instead of: *"You were rejected"* It says: *"If your credit score were 650 instead of 580, you would have been approved."*

This was a big deal because:

- It works on **any** black-box model — no need to open it up
- It's legally defensible and easy to communicate to a non-technical user

🚧 Research Gap (What's Missing?)

The suggestions are often impossible or unrealistic.

The math finds the "nearest point" in feature space — but feature space is abstract. The algorithm doesn't know or care about:

- ❌ **Causal constraints** — e.g., it might suggest "reduce your age by 3 years" — impossible!
- ❌ **Real-world feasibility** — it might suggest an income jump from ₹20k to ₹1 lakh in one month — unrealistic
- ❌ **Feature dependencies** — if income increases, tax bracket also changes, which affects net savings, etc. These cascading effects are ignored.
- ❌ Still no causal mechanism — it's finding the nearest counterfactual statistically, not causally

> **Simple analogy:** The GPS finds the shortest path to your destination, but that path goes through a river. It doesn't know rivers exist. Your project's GPS checks for rivers.

[5] Causal Recourse — The "Fix It the Right Way" Paper

📄 **Full Title:** Algorithmic Recourse: From Counterfactual Explanations to Interventions 👨‍🔬 **Authors:** Amir-Hossein Karimi, Bernhard Schölkopf, Isabel Valera 📅 **Year:** 2021 | 🏛️ **Published at:** ACM FAccT 2021

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/2002.06278
🏛️ ACM Official Page	https://doi.org/10.1145/3442188.3445899

🔬 What Did They Research?

This paper is the **most theoretically advanced** of the six, and the closest in spirit to your project. The researchers asked:

> *"Counterfactual explanations tell you what to change — but they ignore cause and effect. What if we used real causal models to generate recourse that respects how the world actually works?"*

💡 What Did They Find / Build?

They formally established the difference between:

- **Counterfactual recourse** — "what would have to be different" (ignores causality)
- **Interventional recourse** — "what should you actively *do*" (respects causality)

Using **Structural Causal Models (SCMs)** and **Pearl's do-calculus**, they showed mathematically that:

> If you change feature A, and A causally affects feature B, then B will also change as a downstream effect — and your recourse plan must account for that.

Simple analogy:

> The old approach says "increase your savings by ₹50,000." But it ignores that your savings depend on your income and expenses. The causal approach says "to increase savings by ₹50,000, you need to either earn ₹8,000 more per month OR cut ₹4,000 in expenses AND stop 2 credit card payments."

This is **real-world aware** recourse.

🚧 Research Gap (What's Missing?)

Theoretically brilliant, but not practically deployable.

- ❌ **Too computationally expensive** — SCM-based optimization is slow for high-dimensional datasets with many features
- ❌ **No user interface** — pure research output; no dashboard, no tool anyone can actually use
- ❌ **Assumes you already know the causal graph** — in practice, learning the causal graph from data is a hard, separate problem they don't fully address
- ❌ **Tested only on small, toy datasets** — not proven at enterprise scale

> **This is the paper your project directly builds upon and improves** — you take their theoretical foundation, make it computationally feasible, and wrap it in a real decision-support interface.

[6] DiCE — The "Give Me Multiple Options" Paper

📄 **Full Title:** Explaining Machine Learning Classifiers through Diverse Counterfactual Explanations
👨‍🔬 **Authors:** Ramaravind K. Mothilal, Amit Sharma, Chenhao Tan 📅 **Year:** 2020 | 🏛️ **Published at:** ACM FAT* 2020 (by Microsoft Research)

Link	URL
📥 Free PDF (arXiv)	https://arxiv.org/abs/1905.07697
🏛️ ACM Official Page	https://doi.org/10.1145/3351095.3372850
💻 GitHub (Microsoft)	https://github.com/microsoft/DiCE

🔬 What Did They Research?

The researchers (from Microsoft Research) noticed a problem with existing counterfactual tools like Wachter et al.:

> *"They only give you ONE suggestion. But people are different — some can change their income, others can reduce debt. Why not give multiple diverse options and let the user pick?"*

💡 What Did They Find / Build?

They built **DiCE** (Diverse Counterfactual Explanations) — a tool that generates **multiple counterfactual suggestions** that are as different from each other as possible, so the user has real choices.

Simple analogy:

> Instead of a GPS giving you one route, DiCE gives you 3 routes:
>
> - Route 1: Fastest (highway)
> - Route 2: No toll roads
> - Route 3: Most scenic You pick based on what works for you.

For a loan rejection, DiCE might show:

- Option A: Increase income by ₹12,000/month
- Option B: Pay off ₹40,000 in debt + improve credit score by 50 points
- Option C: Reduce loan amount requested by 30%

DiCE works on **any black-box model** and was widely adopted because it was open-sourced by Microsoft.

🚧 Research Gap (What's Missing?)

Diverse options, but still causally blind.

- ❌ **No causal model** — DiCE treats features as independent. It doesn't know that income affects savings, or that credit utilization affects credit score.
- ❌ **Causally impossible suggestions** — it might suggest "reduce your age by 2 years" or "change your employment type" to something that isn't possible for the user's situation
- ❌ **Diversity ≠ Feasibility** — diverse options are great, but if all of them violate real-world causal constraints, diversity adds no value

- ❌ **No real actionability filter** — doesn't distinguish between features the user *can* change vs. *cannot* change with any nuance beyond a simple binary flag

> **Simple analogy:** DiCE gives you 3 GPS routes — but all 3 go through the river it doesn't know about.

🧩 How All 6 Papers Connect (The Story Arc)

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SHAP (2017)          → "We can explain any model's predictions!"
    ↓ Gap: Only correlation, no causation
LIME (2016)          → "We can explain individual predictions locally!"
    ↓ Gap: Unstable, still correlational, no recourse
Wachter (2017)       → "We can tell people what would have changed the outcome!"
    ↓ Gap: Suggestions are causally infeasible (age, impossible jumps)
Ustun (2019)         → "We can make recourse actionable and cost-aware!"
    ↓ Gap: Only works for simple linear models
DiCE (2020)          → "Let's give multiple diverse recourse options!"
    ↓ Gap: Diverse but still causally blind, violates real-world constraints
Karimi (2021)        → "Causal SCMs give true interventional recourse!"
    ↓ Gap: Theoretically perfect but no UI, too slow, not deployed

YOUR PROJECT (2024)  → Combines causal SCM (Karimi) + actionability filtering
(Ustun)
                       + dashboard interface + scalable implementation
```

📝 One-Liner Gap Summary (For Viva)

Paper	What They Did	What's Missing
SHAP	Unified feature importance for any model	Correlation, not causation
LIME	Local explanation for single prediction	Unstable, not causal, no recourse
Ustun et al.	Actionable recourse with cost constraints	Only works for linear models
Wachter et al.	Counterfactual "what-if" explanations	Causally infeasible suggestions
Karimi et al.	Causal SCM-based recourse theory	No UI, not scalable, theoretical only
DiCE	Diverse multiple recourse options	No causal constraints, blind to feature dependencies

TIP

In your viva, if asked *"why didn't you just use SHAP?"* — say: *"SHAP measures correlation, not causation. In high-stakes decisions, a doctor or loan officer needs to know what actually caused the outcome, not just what was statistically associated with it — which is why we used Structural Causal Models instead."*